

Very brief MatLab brush up
Only the very basics

Agenda

- Scripts and functions
- Getting data in and out of MatLab
- Plotting graphs
- Basic vector and matrix operations
- Programming exercises on matrix operations relevant to the course

Structure of MatLab programs

Script: a file of commands – does not contain input and output parameters

May prompt for input or read from file – similar for output

Function: can take input and output parameters

```
function [out1, out2] = test (in1, in2);
```

- Both scripts and functions can call functions
- Functions may also be called from command line in command window
- Functions are good if
 - you need to do the same thing multiple times
 - to structure your program into manageable pieces

Getting data in and out of MatLab

Open these files in MatLab, run them and study how they work

Prompt for input from keyboard, display result in command window

- Simple_input_output.m
 - Simple output methods often useful when debugging

Read and write variables from and to text files

- File_input_output.m
 - Note the basic format checking (can be more advanced)
 - What is the error in the input file?

Plots

Plot 2D graph

- Niceplot.m
 - Include more and more code to see its effect
 - Note how the size and colour of lines and markers are changed
- Plotimage.m
 - An image is a matrix where each pixel has a colour
 - Note how the colour scale may be changed
 - Why do you only get one figure even when plotting C three times?
 - What can you do to get three figures?

Vectors and matrices

The very basics

- Matrix_vector_basics.m
 - Kind of multiple choice – which code line does the job?
 - Or what is the difference between two commands?

Managing memory and runtime – run to see the effect of runtime optimisation

- Preallocation.m
- Matrix_operations_runtime.m
 - MatLab code is interpreted, i.e. slow compared to e.g. compiled C
 - Simple runtime optimisation may pay off – but not always possible

Programming exercises

Exercise 1:

- Program a number of matrix operations using the built-in functions
- Also an exercise in using MatLab Help function

Exercise 2:

- 3 dimensional arrays

Example solutions are given

Note: There are many ways to solve this